Industrial Automation

This is the corrected engineering textbook edition for Industrial Automation 5042. It is not a school notebook. It links power semiconductor devices, converter circuits, industrial drives, heating, UPS systems and PLC ladder programs using diagrams, waveform thinking, field examples, solved numerical problems and answer-writing patterns.

COURSE CODE 5042	CREDITS 4
PERIODS 60	TYPE Program Core / Elective

□ lesson □□□ short note □□□□. Device symbol □□□□□□ □□□□□□□□ □□□ □□□□ □□□□□□ □□□□□□□□. Gate drive, current path, waveform, protection, industrial load, PLC sequence □□□□□□ □□□□□□□□□ □□□□□□□□□□□□□.

COURSE BLUEPRINT

What this handbook is expected to teach

This section converts the dull syllabus table into a usable engineering route: prerequisite knowledge, outcome target, field meaning, diagrams to draw, and problems to solve.

Official objectives

- ✓ Understand the structure, concepts and applications of power semiconductor devices.
- ✓ Explain how controlled rectifiers, inverters, cycloconverters and choppers control industrial power.
- ✓ Connect power electronics to drives, heating, welding and UPS systems.
- ✓ Develop PLC ladder programs for real-time automation applications.

Prerequisite stack

- ✓ Semiconductor Physics from Basic Electronics 2041.
- ✓ Amplifiers and Oscillators from Electronic Circuits 3043.
- ✓ Electrical fundamentals from Electric Circuits and Networks 3041.
- ✓ Logic circuits from Digital Electronics 3044.

How to use

1. Read the concept explanation.
2. Study the diagram or waveform.
3. Solve the worked example without looking.
4. Write the expected-answer format.

CO	Outcome	Hours	Engineering meaning
CO 1	Illustrate structure and operation of power semiconductor devices.	16	Choose and explain MOSFET, IGBT, SCR, DIAC, TRIAC, triggering and commutation.
CO 2	Explain controlled rectifiers, inverters, cycloconverters and choppers.	16	Read converter circuits, current path, firing angle and waveform effect.
CO 3	Illustrate industrial applications of power semiconductor devices.	13	Use converters in drives, heating, welding and UPS.
CO 4	Develop PLC ladder program for an application.	13	Convert process sequence into safe ladder logic with inputs, outputs and interlocks.

REDESIGNED SYLLABUS STRUCTURE

Module map with diagrams, applications and problem targets

The old plain syllabus table is not enough. This map tells what to draw, what to explain, where the marks come from, and how the concept is used in industry.

Module	Core topics	Must draw / visualize	Must solve / apply	Hours
M1: Power semiconductor devices, SCR triggering and commutation	Power MOSFET, IGBT, SCR, DIAC, TRIAC, triggering and commutation.	MOSFET symbol, IGBT symbol, SCR PNPN structure, two-transistor analogy, DIAC and TRIAC symbol, commutation block sketches.	Device selection, SCR current terms, gate triggering comparison.	16
M2: Controlled rectifiers, inverters, cycloconverters and choppers	Single phase controlled rectifiers, AC power control, inverter circuits, dual converter, cycloconverter and chopper.	Half wave controlled rectifier, full bridge, freewheeling diode, inverter block, chopper waveform.	Firing angle effect, duty cycle output voltage, step-down and step-up chopper problems.	16
M3: Industrial applications of power semiconductor devices	AC and DC drives, speed control, industrial heating, resistance welding, online and offline UPS.	AC drive block, DC drive block, induction heating setup, online UPS block, offline UPS block.	Drive comparison table, V/f control explanation, UPS selection question.	13
M4: PLC architecture, instruction set and ladder programs	PLC architecture, advantages, applications, bit, timer, counter, compare, move, math, program control and ladder applications.	PLC scan cycle, input-output table, ladder rungs for start stop, conveyor, level control and traffic signal.	Develop ladder program from process sequence and add interlocks.	13

Power semiconductor devices, SCR triggering and commutation

Industrial automation depends on reliable high-power switches. A power device is not only a symbol in the textbook; it is a gate-drive problem, a heat problem and a protection problem.

Textbook explanation

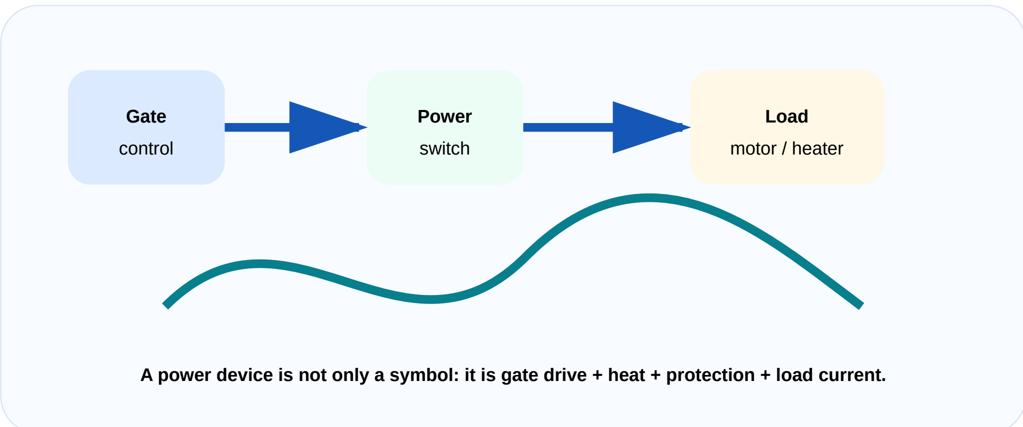
Power MOSFET is a voltage-controlled majority carrier device. It is fast and suitable for low-to-medium voltage choppers, SMPS and DC control. Its conduction loss is mainly decided by RDS(on).

IGBT combines MOS gate control and bipolar conduction. It is widely used in VFDs, UPS inverters and medium-power drives because it handles higher voltage and current effectively.

SCR is a controlled rectifier with PNPN structure. Once triggered, it latches ON and cannot be turned OFF by removing gate signal. Current must fall below holding current or forced commutation must be applied.

DIAC is a bidirectional trigger device. TRIAC is a bidirectional AC power control device used in heater control, fan regulators and lamp dimmers.

> SCR gate pulse ON □□□□□□□□ □□□□□□□ □□□□□□□□□□□□□□□□. OFF □□□□□□□□ holding current □□□□□ □□□□□ current □□□□□□□□□□□ □□□□□□□□□□□ commutation circuit □□□□.



Power switching device as gate control plus load current path.

Engineering subtopics

Power MOSFET

High-speed voltage-controlled switch.

- ✓ Gate, drain and source terminals.
- ✓ Excellent for choppers and SMPS.
- ✓ Low gate current but gate capacitance must be driven.

IGBT

High-power switch used in inverters and VFDs.

- ✓ Gate, collector and emitter terminals.
- ✓ Better for high voltage power control.
- ✓ Needs proper gate resistor and heat sink.

SCR

Controlled rectifier that latches after triggering.

- ✓ Anode, cathode and gate.
- ✓ Holding current and latching current are key terms.
- ✓ Used in controlled rectifiers and AC controllers.

Triggering

Methods that start conduction.

- ✓ Gate triggering is normal method.
- ✓ R, RC and UJT triggering shape firing angle.
- ✓ dv/dt and thermal triggering may be unwanted.

Commutation

Turning OFF a conducting SCR.

- ✓ Natural commutation in AC circuits.
- ✓ Forced commutation in DC circuits.
- ✓ Class A to F methods are asked in exams.

DIAC and TRIAC

Bidirectional AC control devices.

- ✓ DIAC often triggers TRIAC.
- ✓ TRIAC controls both half cycles.
- ✓ Snubber is important with inductive load.

Important formulas / rules

MOSFET conduction loss: $P_{cond} = I^2 R_{DS(on)}$

SCR turns OFF when I_A falls below I_H for required turn-off time.

dv/dt false triggering happens due to junction capacitance current.

Common mistakes

- ✓ Thinking gate can turn SCR OFF.
- ✓ Not mentioning holding current and latching current.
- ✓ Drawing TRIAC as a simple diode instead of bidirectional thyristor.
- ✓ Ignoring snubber requirement for inductive loads.

Worked examples inside this module

Why is IGBT preferred in VFD inverter power stage?

A VFD needs switching of high DC bus voltage and motor current. IGBT provides easy voltage gate control and better high-voltage current handling than ordinary MOSFET, so it is widely used in inverter bridges.

Explain SCR latching with two transistor analogy.

The PNP and NPN equivalent transistors feed each other. A small gate current increases one transistor current, which drives the other transistor. This regenerative action keeps both transistors ON after triggering.

Controlled rectifiers, inverters, cycloconverters and choppers

Converter circuits are the centre of industrial power control. The key is to follow current path and waveform change, not to memorize circuit names only.

Textbook explanation

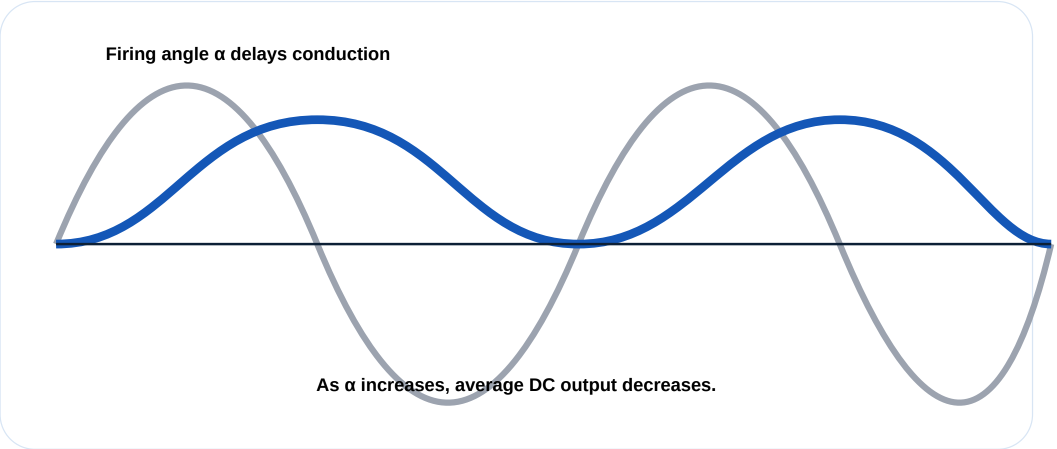
Controlled rectifier converts AC to controlled DC by delaying SCR firing angle. Increasing firing angle reduces the average output voltage.

With R load, current follows voltage. With RL load, current continues due to inductance. A freewheeling diode gives current a safe path and improves load current continuity.

Inverter converts DC to AC. Series and parallel inverters illustrate commutation and waveform formation.

Chopper converts fixed DC into variable DC using switching. Cycloconverter directly changes AC frequency using controlled devices.

> Converter answer □□□□□□□□□□□ circuit □□□□□□ □□□□□□□□. Current path, firing angle, waveform, load effect, application □□□□□□ □□□□□□□□□.



Firing angle and output waveform of controlled rectifier.

Engineering subtopics

Half-wave controlled rectifier

One SCR controls positive half cycle.

- ✓ Simple circuit.
- ✓ Used for low power control.
- ✓ Average output depends on firing angle.

Full-wave controlled rectifier

Provides better DC output using both half cycles.

- ✓ Bridge or midpoint connection.
- ✓ R and RL load waveforms differ.
- ✓ Used in DC drives and chargers.

Freewheeling diode

Inductive load current path during OFF period.

- ✓ Reduces voltage spikes.
- ✓ Improves current continuity.
- ✓ Protects power device.

Inverter

DC to AC converter.

- ✓ Used in UPS and drives.
- ✓ Series and parallel types are syllabus topics.
- ✓ Output waveforms must be drawn.

Cycloconverter

Direct AC to AC frequency converter.

- ✓ Useful in low-speed high-power drives.
- ✓ Can be step-up or step-down frequency concept.
- ✓ Output is made from supply segments.

Chopper

Fixed DC to variable DC.

- ✓ Duty cycle controls output.
- ✓ Step-up and step-down types.
- ✓ Used in DC motor speed control.

Important formulas / rules

Half-wave controlled rectifier: $V_{avg} = (V_m/2\pi)(1 + \cos\alpha)$

Full controlled bridge continuous mode: $V_{avg} = (2V_m/\pi)\cos\alpha$

Step-down chopper: $V_o = D\ V_s$

Step-up chopper: $V_o = V_s/(1-D)$

Common mistakes

- ✓ Forgetting freewheeling diode in RL load answer.
- ✓ Writing chopper as AC converter. It is DC to DC.
- ✓ Not showing waveform after firing angle delay.
- ✓ Confusing inverter and rectifier direction of conversion.

Worked examples inside this module

A 220 V DC supply feeds a step-down chopper with duty cycle 0.4. Find output voltage.

$V_o = D V_s = 0.4 \times 220 = 88$ V.

What happens when firing angle increases in a controlled rectifier?

The SCR conducts later in the cycle. The conduction period reduces and average DC output voltage decreases.

Industrial applications of power semiconductor devices

This module links the devices and converters to real machines. The important question is what is controlled: voltage, current, frequency, torque, speed or heat.

Textbook explanation

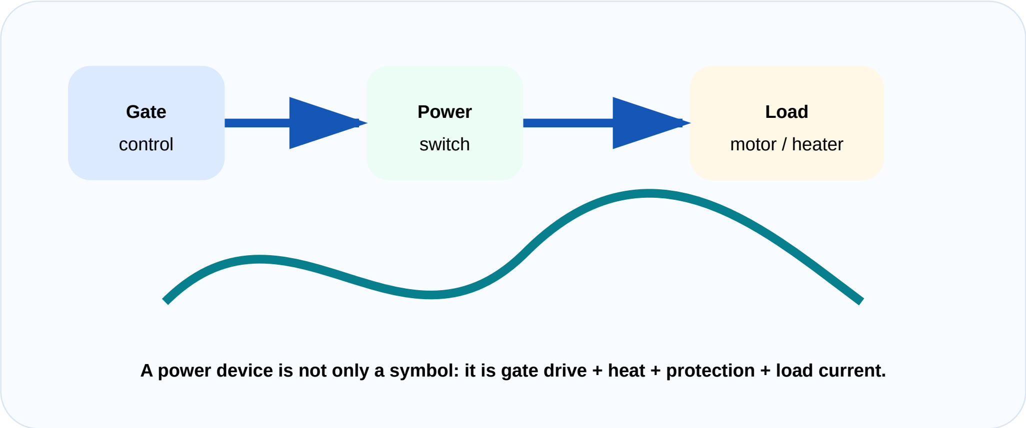
DC drives control speed by armature voltage, field control or resistance method. They offer good torque but need brush maintenance.

AC drives are now common because induction motors are rugged and VFDs provide smooth speed control using VVVF principle.

Induction heating uses induced eddy currents. Dielectric heating uses high-frequency electric field in insulating materials. Resistance welding uses heat generated at joint resistance.

Online UPS supplies load through inverter continuously. Offline UPS supplies load from mains normally and changes to inverter during failure.

Industrial application answer-□ principle, block diagram, merits, applications □□□□□□□ □□□□□□□□□. Example □□□□□□□□□□□□ answer weak □□□□□□□□□□□.



Power electronics applications from converter to load.

Engineering subtopics

AC vs DC drives

Comparison based on motor, control, maintenance and application.

- ✓ AC drive uses induction or synchronous motor.
- ✓ DC drive gives simple torque control.
- ✓ Modern industry prefers VFD for many loads.

VVVF control

Voltage and frequency are changed together.

- ✓ Maintains air-gap flux.
- ✓ Smooth speed control.
- ✓ Used in conveyors, pumps, fans and escalator drives.

Induction heating

Eddy current heat generation.

- ✓ Fast localized heating.
- ✓ Used for hardening and melting.
- ✓ No direct flame needed.

Dielectric heating

Heating of insulating material by electric field.

- ✓ Used for wood, plastic and rubber.
- ✓ Uniform internal heating.
- ✓ High-frequency source required.

Resistance welding

Heat at joint due to high current and contact resistance.

- ✓ Spot, seam and projection welding.
- ✓ Needs pressure and current control.
- ✓ No filler material required.

UPS systems

Backup power with rectifier, battery and inverter.

- ✓ Online has no transfer interruption.
- ✓ Offline is simpler and cheaper.
- ✓ Line conditioning differs.

Important formulas / rules

> Drive control idea: speed is controlled by voltage, frequency or field depending on motor type.

> Induction heating heat source: eddy current I^2R loss in workpiece.

> UPS backup time depends on battery capacity, load power and inverter efficiency.

Common mistakes

- ✓ Writing only definitions without industrial examples.
- ✓ Confusing online and offline UPS energy path.
- ✓ Not mentioning V/f in induction motor speed control.
- ✓ Ignoring protection and feedback in drives.

Worked examples inside this module

Why is VFD preferred for conveyor speed control?

VFD changes frequency and voltage supplied to an induction motor, giving smooth acceleration, speed setting, overload protection and better efficiency than mechanical speed control.

Which UPS is better for critical automation controller?

Online UPS is better because the load is continuously supplied by inverter, giving better power conditioning and no transfer delay.

PLC architecture, instruction set and ladder programs

PLC programming is not drawing contacts randomly. A correct ladder starts from process requirement, input-output list, safety interlock and sequence table.

Textbook explanation

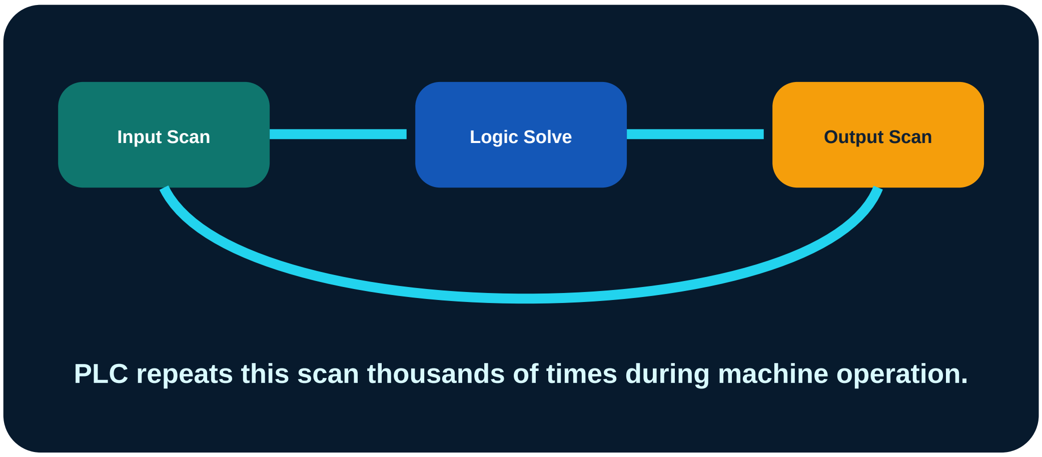
PLC reads field inputs, executes stored logic and updates outputs repeatedly. This is called scan cycle.

Bit instructions implement basic relay logic. Timers, counters, compare, move and math instructions make real process control possible.

A ladder program must include stop, overload, emergency and feedback conditions where required. Unsafe ladder logic may pass a classroom test but fail in industry.

For every PLC answer, write input table, output table, rung logic and explanation.

> PLC ladder program □□□□□□□□□□□□□ □□□□□□□ input, output, sequence, interlock □□□□□□□ table □□□□□□□. Safety contact □□□□□□□□□ ladder real machine-□ □□□□□□□□□□□.



PLC scan cycle and logic flow.

Engineering subtopics

PLC architecture

CPU, memory, power supply, input module, output module and communication.

- ✓ Input scan reads sensors.
- ✓ Program scan solves logic.
- ✓ Output scan updates actuators.

Bit instructions

NO, NC, coil, set and reset.

- ✓ Basic ON-OFF control.
- ✓ Start stop latch.
- ✓ Fault interlocks.

Timers and counters

Delay and event counting.

- ✓ TON for ON delay.
- ✓ Counter for object counting.
- ✓ Reset is necessary.

Compare and move

Data-based decisions.

- ✓ Level threshold.
- ✓ Batch count comparison.
- ✓ Preset transfer.

Math instructions

Simple arithmetic operations.

- ✓ Totalizer.
- ✓ Speed or quantity calculation.
- ✓ Recipe calculation.

Real-time applications

Machine sequence examples.

- ✓ Staircase lamp.
- ✓ Conveyor control.
- ✓ Fluid level control.
- ✓ Traffic signal.

Important formulas / rules

PLC scan = input scan + program scan + output scan + communication housekeeping.

Start-stop latch: STOP and fault contacts in series, START and holding contact in parallel.

Counter logic: sensor pulse increments count; compare count to preset; reset after batch complete.

Common mistakes

- ✓ No stop or overload contact in holding rung.
- ✓ Forward and reverse output allowed together.
- ✓ Counter not reset after batch completion.
- ✓ Using timer without checking feedback.

Worked examples inside this module

Develop a start-stop latch concept.

Place STOP and OVERLOAD normally closed contacts in series. Place START contact in parallel with MOTOR auxiliary contact. Coil is MOTOR. This keeps motor ON after releasing START and stops it during fault.

Conveyor counting logic method.

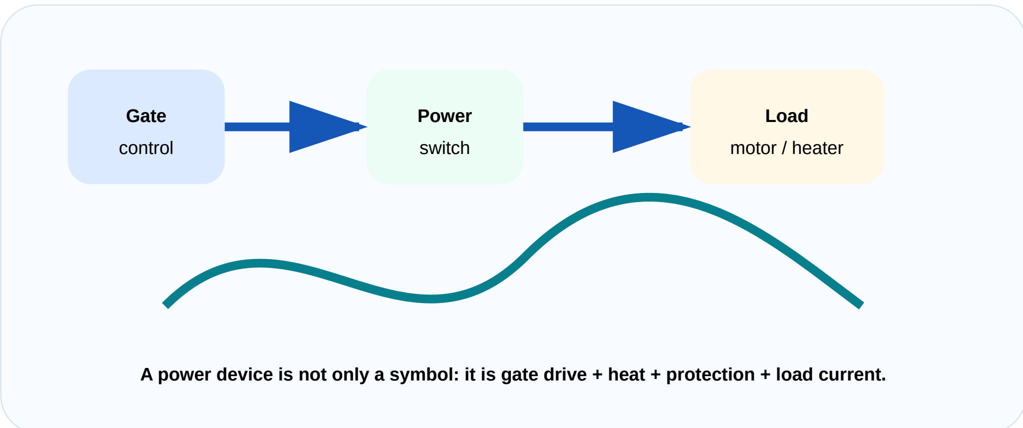
Photo sensor pulse drives CTU counter. When accumulated value reaches preset, batch complete output turns ON. Reset push button resets the counter for next batch.

ILLUSTRATION LAB

Images, block diagrams and waveform thinking

For technical subjects, the diagram is half of the answer. These visuals make the lesson look and work like a textbook instead of a plain note.

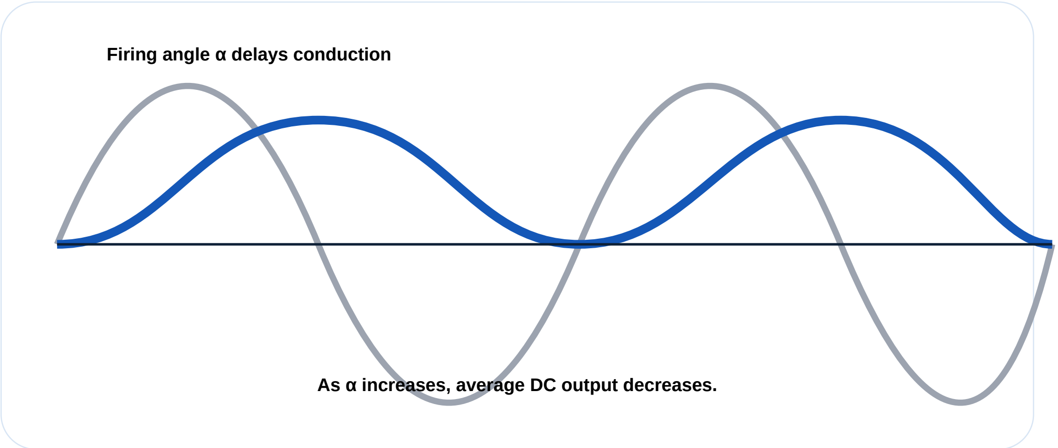
Power device control chain



Gate drive controls a high-power load through semiconductor switch.

Use this thinking for MOSFET, IGBT and SCR questions.

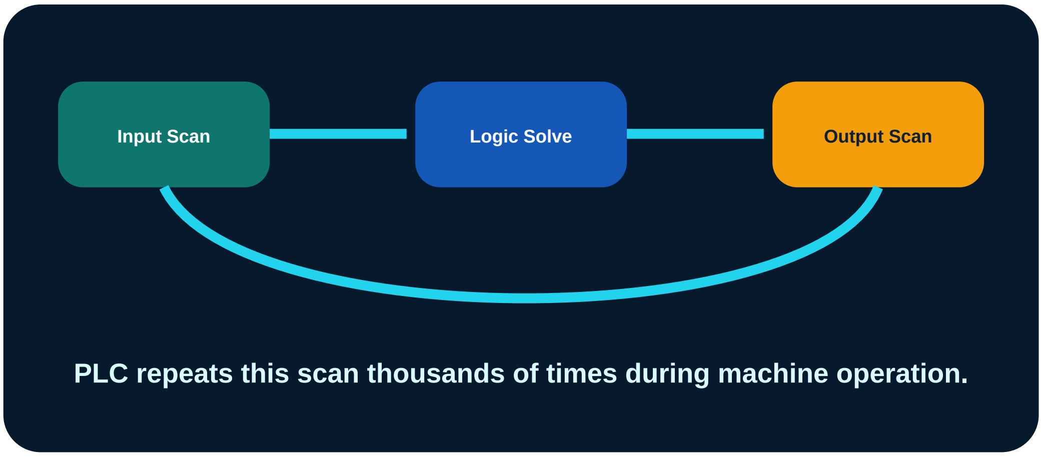
Controlled rectifier waveform



Firing angle controls average DC output.

Always mark α and explain why output decreases as α increases.

PLC scan cycle



PLC reads, solves and updates repeatedly.

This is the base of every ladder program explanation.

Industrial system block idea

ery industrial system has input, processing and outp

Diagram, waveform, block flow and practical meaning should be studied together.

Every industrial system has input, processing and output.

Same concept applies to sensors, PLC and actuator control.

PROBLEM LAB

Solved numerical and design problems

Each problem shows the method, not only the answer. This is the difference between a notebook and an engineering handbook.

Chopper output voltage

Problem: A 250 V DC source feeds a step-down chopper with duty cycle 0.35. Find average output voltage.

1. Identify converter as step-down chopper.
2. Use $V_o = D\,V_s$.
3. Substitute $D = 0.35$ and $V_s = 250$ V.

Answer: $V_o = 87.5$ V

SCR turn-off reasoning

Problem: An SCR is conducting. Gate signal is removed but load current is still greater than holding current. Will SCR turn OFF?

1. SCR is a latching device.
2. Gate removal alone cannot turn it OFF.
3. Turn-off happens only when anode current falls below holding current for turn-off time.

Answer: No. It remains ON.

VFD application

Problem: Why is VVVF control used for induction motor speed control?

1. Induction motor speed depends on supply frequency.
2. Changing only frequency can disturb flux.
3. Changing voltage and frequency together maintains proper flux and gives smooth speed control.

Answer: VVVF gives efficient smooth speed control.

PLC object counting

Problem: A conveyor must stop after 12 objects are detected. State the logic method.

1. Use photo sensor as counter input.
2. Use CTU counter with preset 12.
3. When counter done bit is ON, stop conveyor and turn ON batch complete lamp.
4. Use reset push button to clear count.

Answer: Photo sensor → CTU preset 12 → done bit stops conveyor.

PRACTICE

Expected questions by module

Module 1

Explain construction and working of SCR with V-I characteristics.

Compare power MOSFET and IGBT.

Explain two transistor analogy of SCR.

Explain R, RC and UJT triggering methods.

Describe Class A to Class F commutation techniques.

Explain DIAC and TRIAC with applications.

Module 2

- Explain single phase half wave controlled rectifier with R and RL load.
- Explain full wave controlled bridge rectifier with waveform.
- Explain AC power control using SCR and TRIAC.
- Describe series and parallel inverter circuits.
- Explain dual converter and cycloconverter.
- Explain step-up, step-down and Jones chopper.

Module 3

- Compare AC and DC drives.
- Explain speed control techniques of induction motor.
- Explain induction heating and dielectric heating.
- Explain resistance welding schemes.
- Compare online and offline UPS.
- Write industrial applications of power semiconductor devices.

Module 4

- Draw PLC architecture and explain scan cycle.
- List advantages and applications of PLC.
- Explain bit, timer, counter, compare, move and math instructions.
- Develop ladder program for staircase lamp.
- Develop ladder program for conveyor control and object counting.
- Develop ladder program for fluid level control and traffic signal.

MODEL PAPER

Full practice question paper and answer guide

This model follows the syllabus order and avoids copying official papers.

COURSE 5042 INDUSTRIAL AUTOMATION
Time: 3 Hours Maximum Marks: 100

PART A
1. Define holding current and latching current.
2. State two applications of IGBT.
3. What is freewheeling diode?
4. Define duty cycle.
5. List two applications of TRIAC.
6. What is VVVF control?
7. State two merits of induction heating.
8. What is PLC scan cycle?
9. Name two PLC timer or counter instructions.
10. List two PLC applications.

PART B
11. Explain power MOSFET structure and working.
12. Explain two transistor analogy of SCR.
13. Explain RC and UJT triggering.
14. Explain half wave controlled rectifier with R load.
15. Explain step-down chopper with equation.
16. Compare AC drive and DC drive.
17. Explain online UPS with block diagram.

18. Develop ladder logic for staircase lamp.

PART C
19. Explain SCR V-I characteristics and turn-on methods.
20. Describe Class A to Class F commutation.
21. Explain full wave controlled rectifier with RL load and freewheeling diode.
22. Explain inverter and cycloconverter applications.
23. Explain industrial heating and resistance welding.
24. Draw PLC architecture and explain instruction sets.
25. Develop ladder program for conveyor counting system.
26. Develop ladder program for fluid level control.

Answer writing guide

- ✓ Use diagram plus principle for every power electronics answer.
- ✓ For converter answers include circuit, current path, waveform and application.
- ✓ For PLC answers include input-output table, ladder rung and rung explanation.
- ✓ For application questions include principle, merits, limitations and real industrial use.

REFERENCES

Books and resources

Type	Reference
T1	Industrial Electronics and Control - S K Bhattacharya, S Chatterjee
T2	Programmable Logic Controllers - Frank D Petruzella
R1	Industrial Electronics and Control - Biswanath Paul - PHI
R2	Thyristors Principles and Applications - Ramamoorthy
R3	Modern Power Electronics and AC Drives - Bimal K Bose
R4	Power Electronics - P S Bimbhra
Online	NPTEL 108102145, electronics-tutorials.ws, technologystudent.com, NDL IIT Kharagpur, Electrical4U

Final revision rule: Prepare one diagram, one formula, one application and one common mistake for every major topic.